Module: CMP6046B/7040B Ubiquitous Computing
Assignment: A4: Indoor Localization
(Part of Portfolio of Lab Work A3, A4)

Set by: Dr Min Hane Aung, Min.Aung@uea.ac.uk
Date set: 10 March 2026
Value: 20%

Date due: 30 April 2026 15:00
Returned by: 10 June 2026
Submission: Report submission on Blackboard
Checked by: Dr Xu Zhang

Learning outcomes

Gain experience in collecting WiFi signal strength data at different indoor locations using a smartphone. Understand and implement similarity measures to analyse the proximity of the different locations to relative to each other.

Specification

Overview

<u>This assignment is not done in groups and must be done in your own.</u>

The aim of the assignment is to collect your own data using our pre-built smartphone app (Android only) of WiFi access point information and signal strength. You will save your data onto the phone's local folder system and use the specific file naming convention described in the **A4 Collecting Data Instruction** sheet.

Once all data is collected you will analyse the data in a Python environment in the lab or on your laptops by implementing a similarity metric by following the instructions in the **A4 Data Analysis Instruction** sheet.

Description

<u>Stage 1 – App Installation and Data Collection</u>

The lab issued phones should already have the app installed. However, if you are using your own phone or a lab phone does not have the app please follow the instructions in the **A4 App Installation Instructions** document.

Please note in later model phones (e.g., Samsung S8) you may need to manually set some permission settings to allow the scans and data to be saved. Please see the **A4 Manual Permission Settings in Later Phones** document.

Read and follow carefully the **A4 Collecting Data Instruction** sheet. You will need to go outside of the lab to complete this stage. Please also see the **A4 Collecting Data Tutorial Video.**

If you are unable to collect data it maybe possible to use the supplied reserve data, please check with Dr Aung for permission.

Stage 2 – Data Analysis and Classification

Return to the lab and commence the data analysis part and follow the **A4 Data Analysis Instruction Sheet**. This will be done in either Spyder or Jupyter Notebooks the Python environments. If you are unfamiliar with these please take some time to follow the **Getting Started with Python** sheet.

Stage 3 – User Feedback Design

Using the similarity measures output from Stage 2, design and create a Lo-Fi visualization of the Stage 2 analysis output that will be more intuitive to a non-technical user than the matrix scores. You should consider that the design should be clear and comprehendible but also accurate and informative and importantly it should represent the data.

You can create a design by hand or use drawing tools, all images should be embedded into your final pdf file and not submitted as separate files. If you wish to make of further visualizations tools in Python this is permitted

Relationship to formative assessment

This assignment uses an Android based app that utilizes the smartphone sensor as developed in formative assignment A1 and is similar to the data analyses procedure done in assignment A3 stage 2

Deliverables

Each student will write and submit a PDF report which contains all of the requirements in stages 2 and 3.

This report must be eventually submitted as part of the Portfolio of Coursework in one single combined PDF file.

Resources

Main Materials

A4 App Installation Instructions

A4 Collecting Data Instructions

A4 Data Analysis Instructions

Phone Apps – IndoorLocalizationA4_V1.apk or IndoorLocalizationA4_V2.apk

Analysis Script – wifi_localization.py

(A4 Reserve Data)

A4 Collecting Data Tutorial Video

Additional Materials

Getting Started with Python Document

Sensors Lecture Slides

Signal Processing and Feature Engineering Lecture Slides

Data Science and Machine Learning Lecture Slides

Marking scheme

Activity	Marks	Awarded
Stage 2 (To Do 1 & 2)	6	
Stage 2 (To Do 3 & 4)	6	
Stage 3 Design	8	
	Total	/20

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